

Compile and burn boot

Brief description: The boot stage of the SSD202D chip is also what we often call the bootloader. It is developed and adapted based on uboot2015. A large number of original chip related codes have been added to it, which is very different from the community version. Because SSD202 only supports NANDFLASH / NorFLASH So if there is no boot partition on your development board or the boot partition is accidentally damaged, the system will not be able to start normally, and you need to use a special programming tool to update it.

The programming tool is affected by the supply of the chip market, and the price is very high, so we do not recommend you to operate this part.

If you find that you have problems following this document, you can directly view our pre-recorded hands-on instructional video.

compile bootloader

First, go to the root directory of the boot source code. Before compiling, you need to confirm whether the cross-compilation toolchain and other environments have been successfully configured. If you are not sure, you can refer to [the](#) section Configuring the Cross-compilation Toolchain.

```
make infinity2m_spinand_defconfig
```

After specifying the configuration file, continue to execute the following command to start compiling (according to the server configuration, multi-threaded compilation can be supported)

```
make -j8
```

- Reference execution example

```
book@100ask:~/DongshanPiOne-TAKOYAKI/boot$ make
infinity2m_spinand_defconfig
arch/./configs/.tmp_defconfig:326:warning: override: reassigning to symbol
CMD_FASTBOOT
#
# configuration written to .config
#
book@100ask:~/DongshanPiOne-TAKOYAKI/boot$ make -j16
```

The final generated image file is:

```
u-boot_spinand.xz.img.bin
```

Note: If you need to package the compiled image into the final complete system, you need to copy the generated files to the project directory:

```
cp u-boot_spinand.xz.img.bin project/board/i2m/boot/spinand/u-boot/
```

- Reference execution example

```
book@100ask:~/DongshanPiOne-TAKOYAKI/boot$ cp u-boot_spinand.xz.img.bin  
../project/board/i2m/boot/spinand/u-boot/
```

Program the bootloader

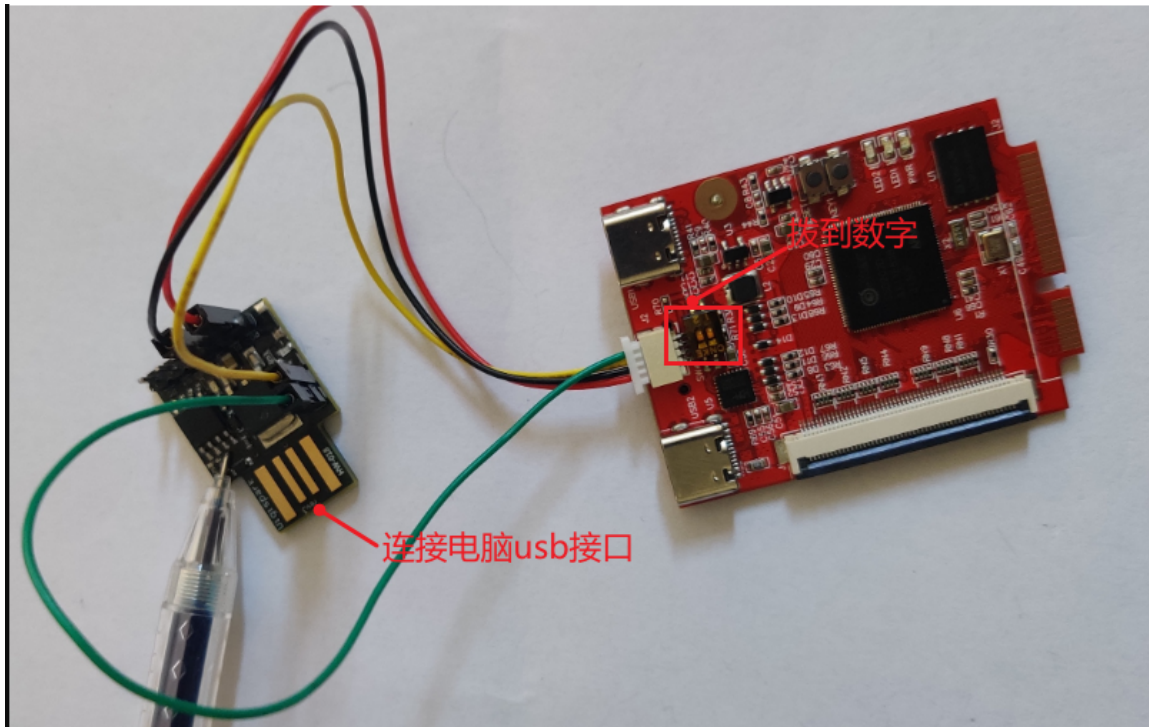
Enter uboot to program

Note: **This programming method is that your development board can still be started normally, and you can enter the uboot command line.** First, copy the compiled image to your sd card by referring to the above operation, then power on the development board, keep pressing the Enter key on the keyboard for about 3 seconds when there is output from the serial port, and then release it to enter the uboot command line, and finally execute Update with the following command.

```
fatload mmc 0:1 0x21000000 u-boot_spinand.xz.img.bin  
  
nand erase.part UBOOT0  
  
nand write.e 0x21000000 UBOOT0 ${filesize}  
  
nand erase.part UBOOT1  
  
nand write.e 0x21000000 UBOOT1 ${filesize}
```

Flash using open source community tools

- For a more detailed description of open source community tools, please see;
- 社区版烧写工具 连线介绍 <https://item.taobao.com/item.htm?id=665728385852>
- 将烧写工具连接好烧录线后，将烧录器和开发板烧写接口连接好，拨码开关切换为烧写模式（拨码开发白色拨码切换到非NO方向）。



之后我们按下主板上 Reset 键，同时执行如下烧写命令，其中第一句话 `sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -e` 是整个 擦除 flash 命令，建议执行。

注意:烧写操作是在ubuntu-18烧写，需要将烧写工具挂载到ubuntu系统内，挂载成功后系统会自动装在驱动模块，此时设备管理器会多出一个 `/dev/i2c-0` 的设备节点，其中烧写命令中 `-l` 参数指定的是烧写文件的长度(大小)，如果你修改了 `u-boot_spinand.xz.img.bin` 文件，请确认大小是否在这个区间内。

```
sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -e

sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -w GCIS.bin

sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -a 0x140000 -l 0x6000 -w IPL.bin

sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -a 0x200000 -l 0x5800 -w
IPL_CUST.bin

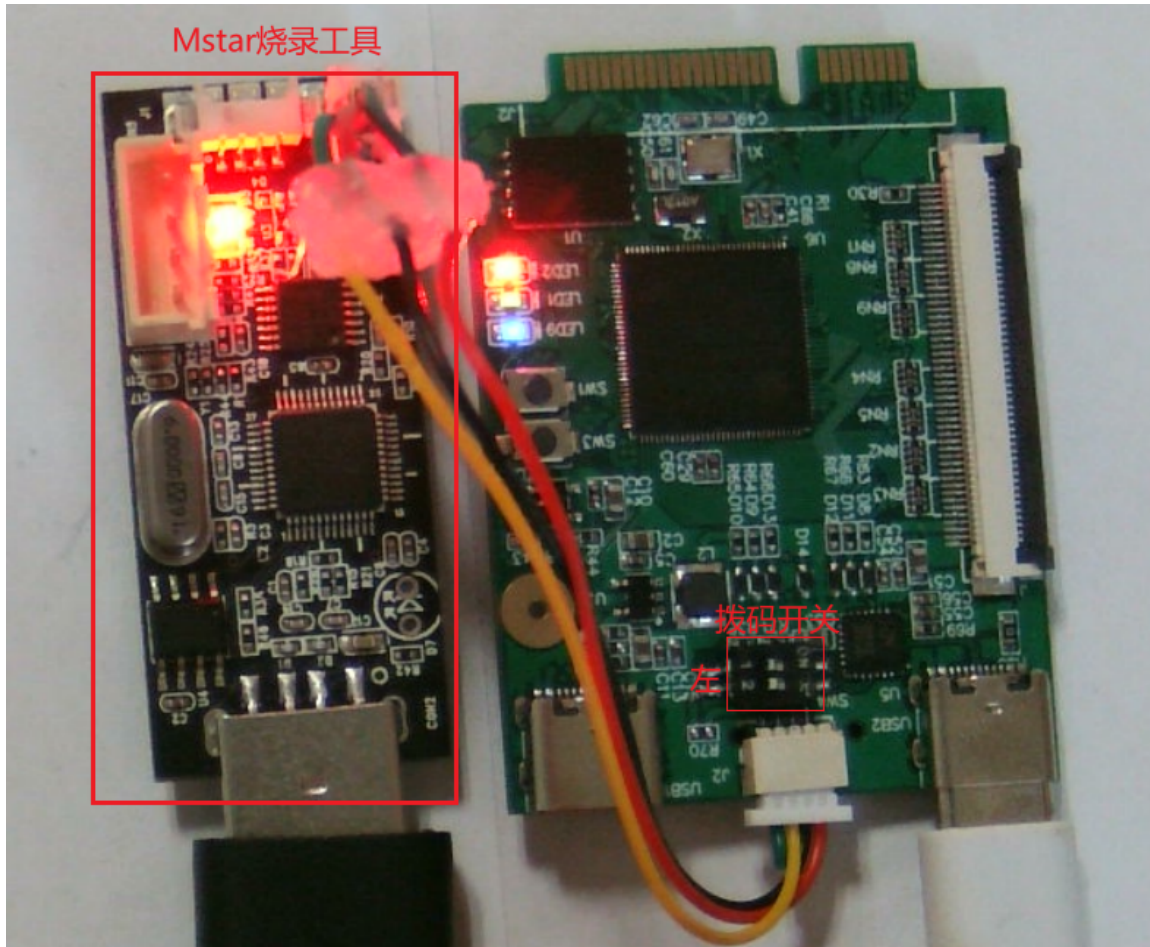
sudo SNANDer -p mstarddc -c /dev/i2c-0:49 -a 0x2C0000 -l 0x3B800 -w u-
boot_spinand.xz.img.bin
```

使用专门的烧写器烧写

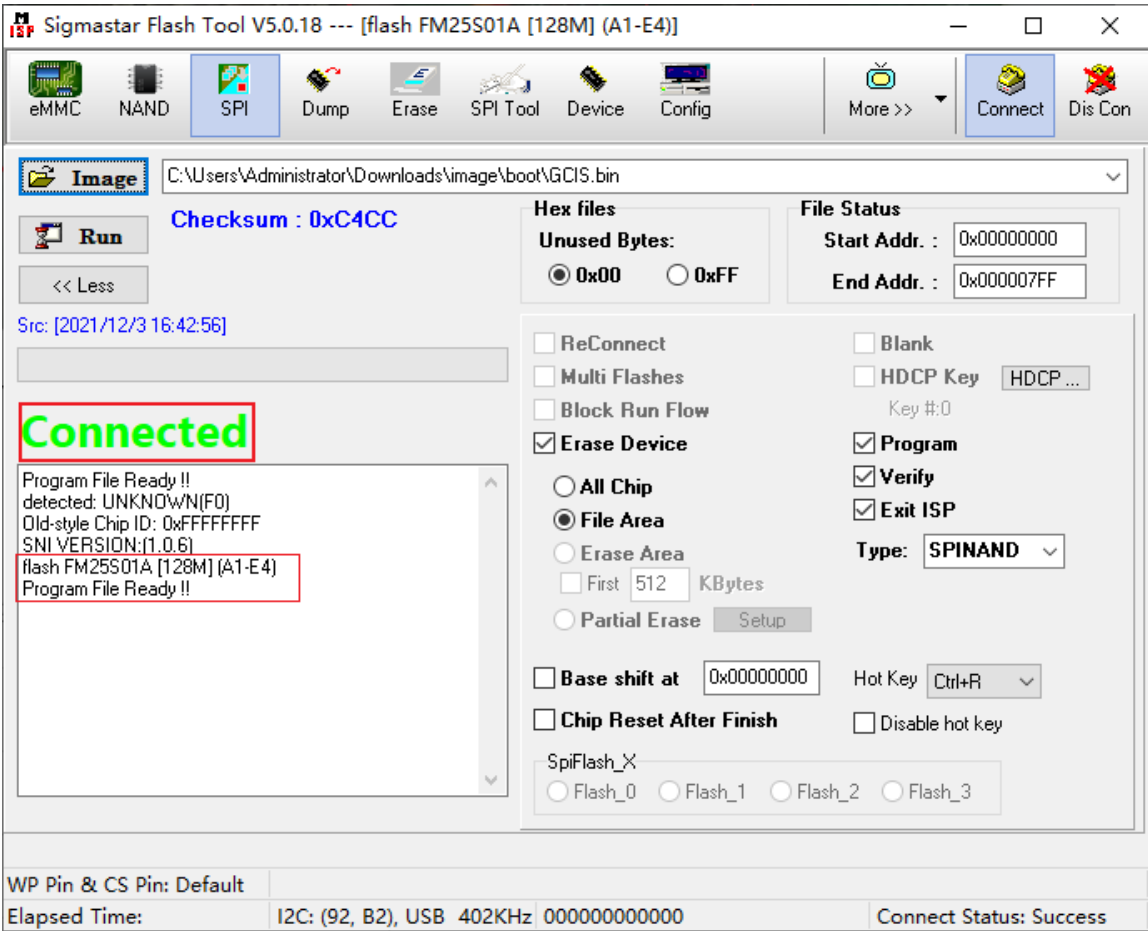
注意：本方式适用于空片烧录或者板子无法进入Uboot控制台使用其它方式烧录的情况。

首先关闭串口调试工具，接下来打开烧写工具：烧写工具下载 [点击跳转](#) 选择合适的下载方式下载。其中烧录器购买链接地址：

同时将烧录器和开发板烧写接口连接好，拨码开关切换为烧写模式(拨码开发白色拨码切换到非NO方向)。



电脑解压下载后的 **ISP_5.0.18.zip** 烧写工具压缩包，打开 **Flash_Tool_5.0.18.exe**，点击 **Connect**，建立连接状态（**Connect**必须确保关闭串口工具，否则会出现争抢串口资源问题）选择Flash Type为Nand Flash。

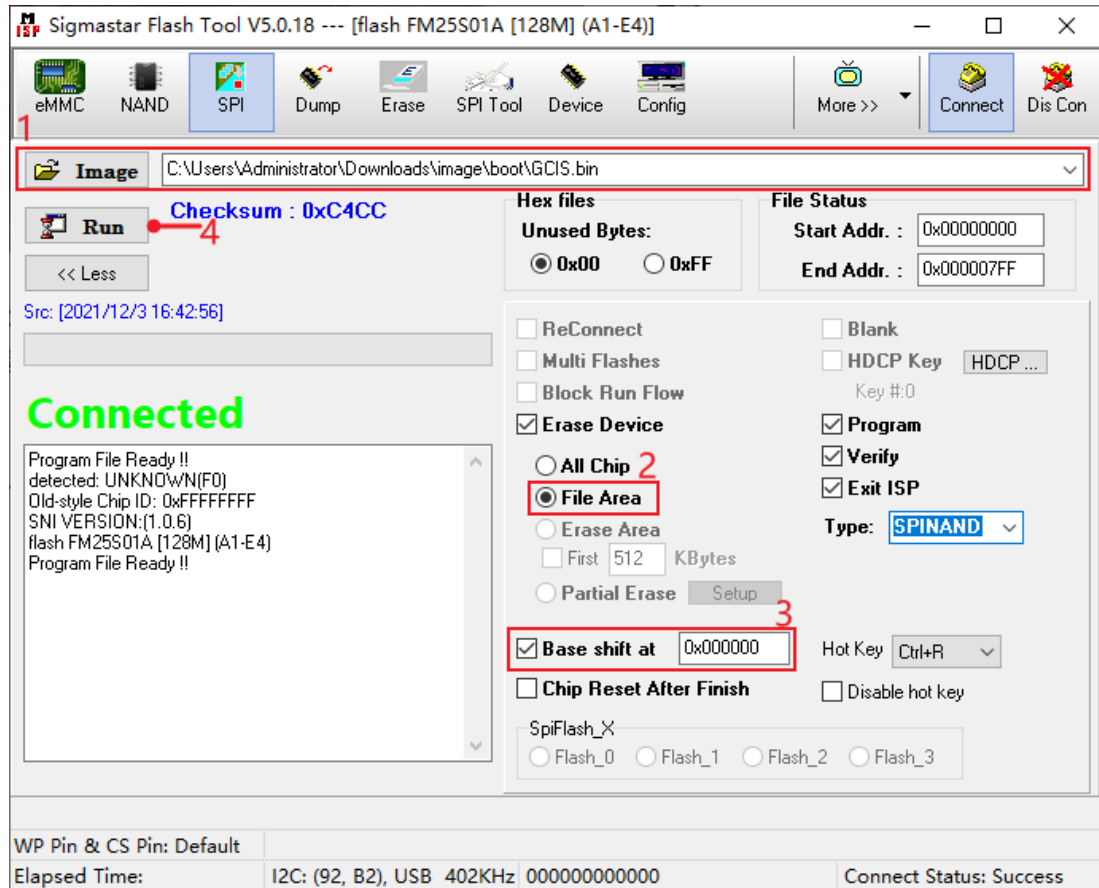


接下来需要根据以下的分区以及分区起始地址，按照以下方法依次烧录分区。其中 GCIS.bin IPL.bin IPL_CUST.bin u-boot_spinand.xz.img.bin 这些文件可以从我们提供提供默认系统镜像中获取 [恢复默认系统](#)。

分区文件	分区起始地址
GCIS.bin	0x000000
IPL.bin	0x140000
IPL_CUST.bin	0x200000
u-boot_spinand.xz.img.bin	0x2C0000

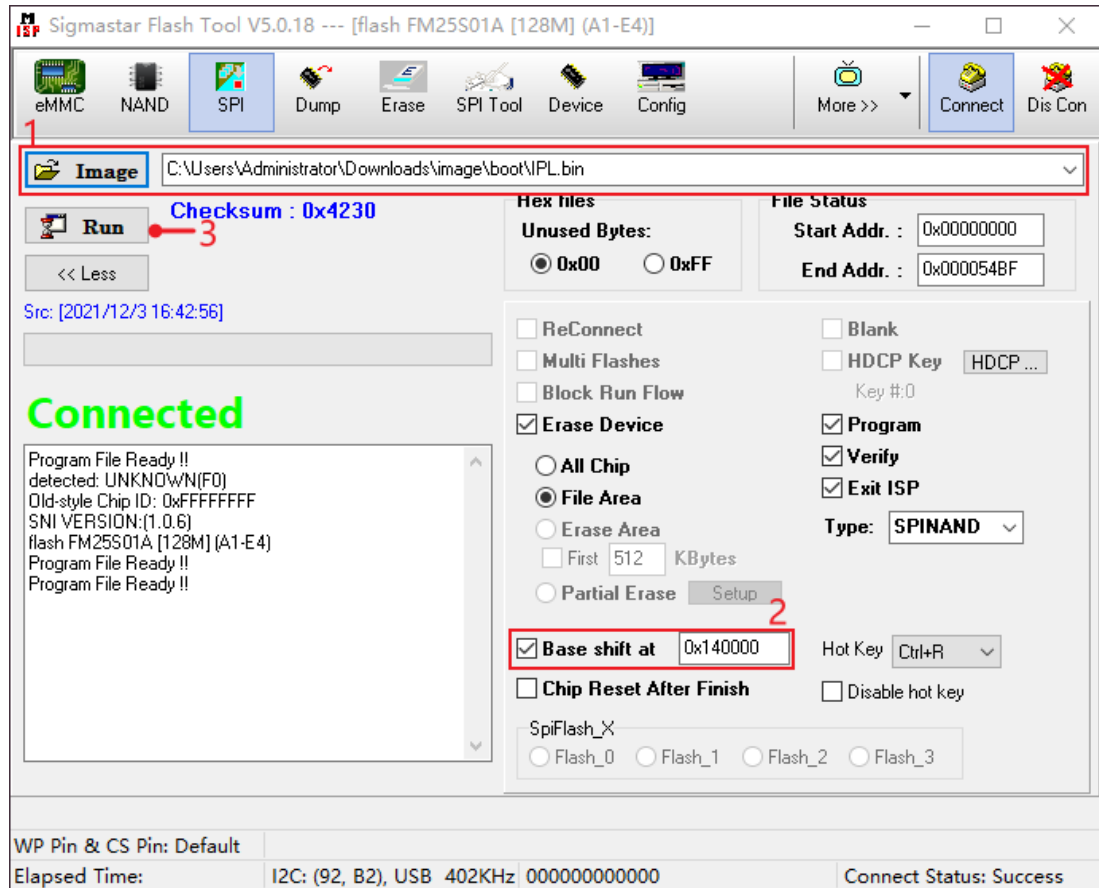
- 烧录 GCIS.bin :参考下图 1 红框选择文件所在位置，(注意这些镜像文件来源于 前面 常见问题章节 更新默认系统部分)。选择完成后，参考下图 2 红框设置烧写地址，配置完成后可以

点击 **箭头3** Run 按钮来开始烧录。等待运行结束，直至提示Pass状态。



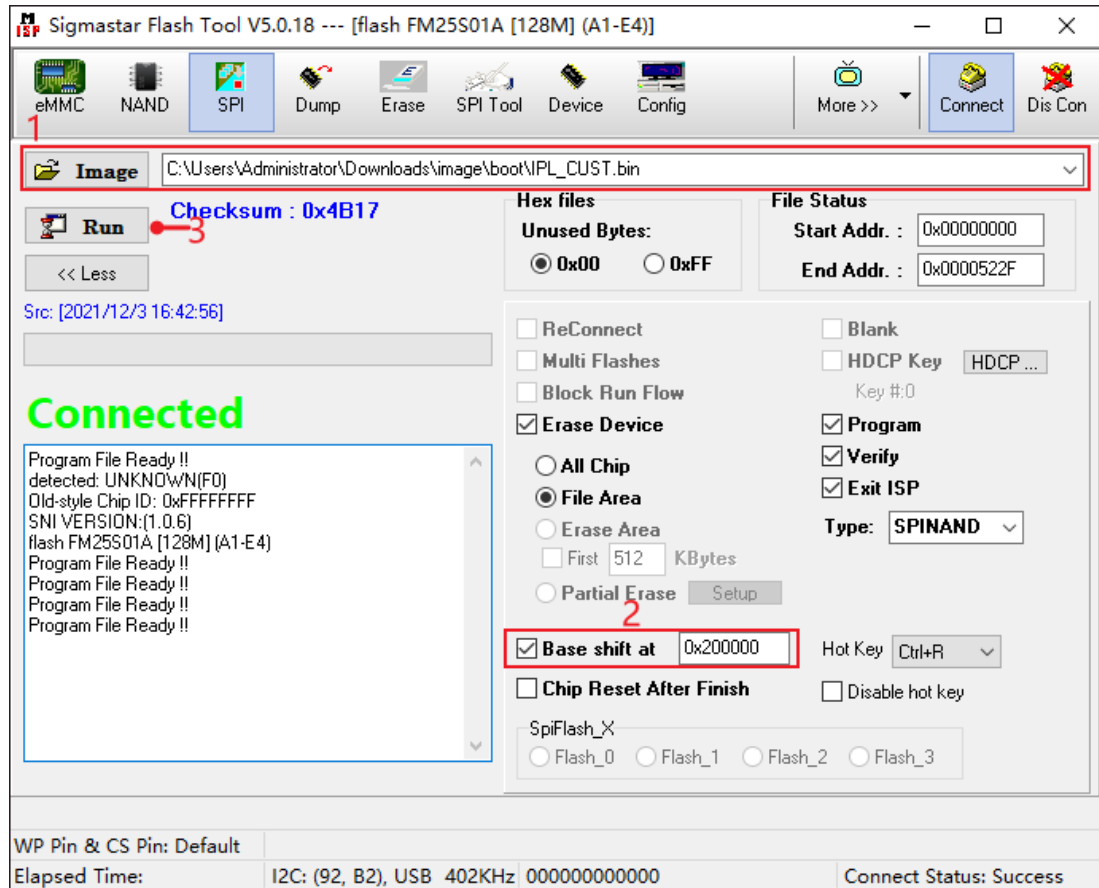
- 烧录 IPL.bin :参考下图**1 红框选择文件所在位置，(注意这些镜像文件来源于 前面 常见问题章节 更新默认系统部分)。选择完成后，参考下图 **2 红框设置烧写地址，配置完成后可

以点击 **箭头3** Run 按钮来开始烧录。等待运行结束，直至提示Pass状态。

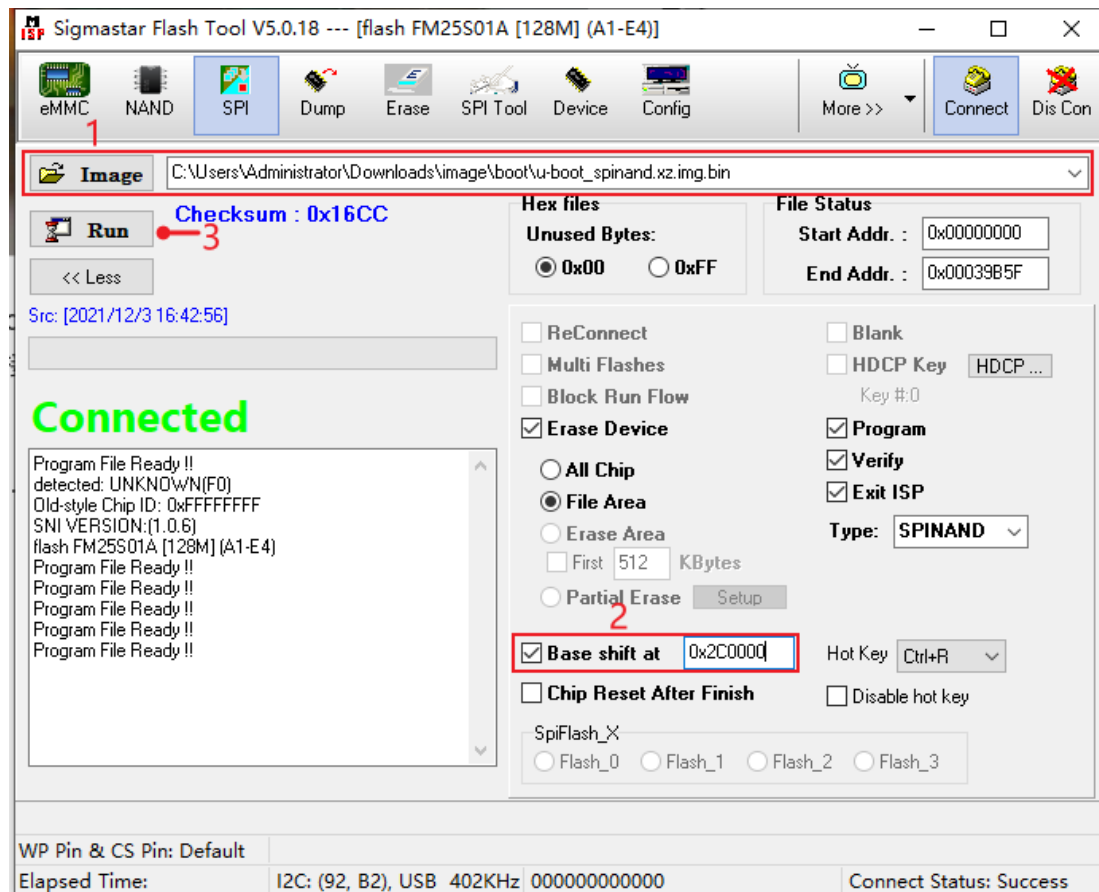


- 烧录 IPL_CUST.bin :参考下图**1 红框选择文件所在位置，(注意这些镜像文件来源于 前面 常见问题章节 更新默认系统部分)。选择完成后，参考下图 **2 红框设置烧写地址，配置完成

后可以点击 **箭头3** Run 按钮来开始烧录。等待运行结束，直至提示Pass状态。



- Burn u-boot_spinand.xz.img.bin : Refer to the following figure **1 red box to select the location of the file, (note that these image files are from the previous FAQ section Update the default system section). After the selection is complete, refer to the red box in the figure below to set the programming address. After the configuration is complete, you can click the **arrow** Run 3 button to start programming.



Click Run and wait for the end of the run until the Pass status is prompted. After burning, execute reset and restart to start normally.